

Functional data analysis

6. Statistical inference for functional data

Alfredas Račkauskas, Jurgita Markevičiūtė

Lecture notes, 2026

Objective are:

- ▶ *one sample inference problems including change point analysis;*
- ▶ *two sample problems;*
- ▶ *functional ANOVA.*

fdANOVA: An R Software Package for Analysis of Variance for
Univariate and Multivariate Functional Data

Outline of the lecture

Point estimation of model parameters

One-Sample Problems for Functional Data

Two sample problems

Methods for Two-Sample Problems

One-Way ANOVA

One-way ANOVA tests

Change point problem

Detection of changes in the mean function

Point estimation of model parameters

- Consider a random functional sample of size n ,

$X_1(t), \dots, X_n(t), \quad t \in \mathcal{I} := [0, 1], \quad \overset{iid}{\sim} \quad Sp(\mu, \gamma)$ for some μ, γ ,
and denote by $X = (X(t), t \in \mathcal{I})$ its generic element.

Throughout we assume that X is $L_2(\mathcal{I})$ random element.

- $\mu = (\mu(t), t \in \mathcal{I})$ is the mean function,

$$\mu(t) = E(X(t)), \quad t \in \mathcal{I};$$

- $\gamma = (\gamma(s, t), s, t \in \mathcal{I})$ is the covariance function,

$$\gamma(t, s) = E[(X(t) - \mu(t))(X(s) - \mu(s))], \quad t, s \in \mathcal{I}.$$

- The covariance operator $\mathcal{K} = \mathcal{K}_X : L_2(\mathcal{I}) \rightarrow L_2(\mathcal{I})$ is defined by

$$\mathcal{K}(f)(t) = \int_{\mathcal{I}} \gamma(t, s) f(s) \, ds, \quad f \in L_2(\mathcal{I}).$$

- When γ has finite trace:

$$\mathrm{tr}(\gamma) := \int_{\mathcal{I}} \gamma(t, t) \, dt < \infty,$$

it has the following singular value decomposition (SVD):

$$\gamma(t, s) = \sum_{r=1}^m \lambda_r \psi_r(t) \psi_r(s),$$

where $\lambda_1, \lambda_2, \dots$ are all the decreasingly ordered positive eigenvalues of γ , $\psi_1, \psi_2, \dots$ are the associated orthonormal eigenfunctions of γ such that

$$\int_{\mathcal{I}} \psi_r^2(t) \, dt = 1, \quad \int_{\mathcal{I}} \psi_r(t) \psi_l(t) \, dt = 0, \quad r \neq l$$

and m is the smallest integer such that when $r > m$, $\lambda_r = 0$.

- Notice that $m = \infty$ when all the eigenvalues are positive.

- Then X has the following Karhunen-Loève expansion:

$$X(t) = \mu(t) + \sum_{r=1}^m \xi_r \psi_r(t),$$

where $\xi_r = \langle X, \psi_r \rangle$, $r = 1, 2, \dots, m$ are uncorrelated,

$$E\xi_r = 0, \quad \text{and} \quad E\xi_r^2 = \lambda_r.$$

- The operator $\mathcal{K}$ has the decomposition

$$\mathcal{K} = \sum_{r=1}^m \lambda_r \psi_r \otimes \psi_r.$$

The SVD of a covariance function and the Karhunen-Loève expansion of a random function plays a central role in FDA.

Estimation of the mean function

- The mean function μ is estimated by the sample mean function $\hat{\mu} = (\hat{\mu}(t), t \in \mathcal{I})$,

$$\hat{\mu}(t) = n^{-1} \sum_{j=1}^n X_j(t).$$

Theorem

- (a) The estimator $\hat{\mu}$ is unbiased:

$$E(\hat{\mu}(t)) = \mu(t), \quad t \in \mathcal{I}.$$

- (b) If $(X_i) \sim GSp(\mu, \gamma)$, then $\hat{\mu} \sim GSp(\mu, n^{-1}\gamma)$,

- (c) If $E(\|X\|^2) < \infty$, (equiv. $\text{tr}(\gamma) < \infty$), then

$$E(\|\hat{\mu} - \mu\|^2) = n^{-1}E(\|X - \mu\|^2).$$

Proof.

(a) Exercise.

(b) If $\xi_1, \dots, \xi_n \sim \mathcal{N}(a, \sigma^2)$, then $n^{-1/2} \sum_{i=1}^n \xi_i \sim \mathcal{N}(a, \sigma^2)$.

(c) We have

$$\begin{aligned} \mathbb{E}(\|\hat{\mu} - \mu\|^2) &= n^{-2} \sum_{i,j=1}^n \mathbb{E}[\langle (X_i - \mu), (X_j - \mu) \rangle] \\ &= n^{-2} \sum_{i=1}^n \mathbb{E}\|X_i - \mu\|^2 \\ &= n^{-1} \mathbb{E}\|X - \mu\|^2 \end{aligned}$$

We used: $\mathbb{E}\langle X_i - \mu, X_j - \mu \rangle = 0$ if $i \neq j$.

Estimation of covariance function

- The covariance function $\gamma(t, s)$ is estimated by its sample counterpart

$$\hat{\gamma}(s, t) = n^{-1} \sum_{i=1}^n (X_i(t) - \hat{\mu}(t))(X_i(s) - \hat{\mu}(s)), \quad s, t \in \mathcal{I}.$$

- Just as in the scalar case, $\hat{\gamma}(t, s)$ is a biased estimator of $\gamma(t, s)$. An elementary verification shows that

$$\mathbb{E}(\hat{\gamma}(t, s)) = \frac{n}{n-1} \gamma(t, s), \quad t, s \in \mathcal{I}.$$

Estimation of covariance operator

- The covariance operator $\mathcal{K}$ is estimated by its sample counterpart

$$\hat{\mathcal{K}} = n^{-1} \sum_{i=1}^n (X_i - \hat{\mu}) \otimes (X_i - \hat{\mu}_n),$$

hence,

$$\hat{\mathcal{K}}h = n^{-1} \sum_{i=1}^n \langle (X_i - \hat{\mu}), h \rangle (X_i - \hat{\mu}), \quad h \in L_2.$$

Estimation of eigenvalues and eigenfunctions

- ▶ The functional principal components (FPC's) are orthonormal eigenfunctions of the covariance operator.
- ▶ The eigenfunctions v_j are defined by

$$\mathcal{K}(v_j) = \lambda_j v_j,$$

so if v_j is an eigenfunction, then so is av_j , for any non-zero scalar a (by definition, eigenfunctions are non-zero functions).

- ▶ The FPC's v_j are normalized (orthonormal by definition), so $\|v_j\| = 1$, but this does not determine the sign of v_j .
- ▶ Thus if $\hat{v}_j$ is an estimate computed from the data, we can only hope that $\hat{c}_j \hat{v}_j$ is close to v_j , where

$$\hat{c}_j = \text{sign}(\langle \hat{v}_j, v_j \rangle).$$

- ▶ When discussing properties of the $\hat{v}_j$ we will always implicitly assume that this sign has been taken into account,
- ▶ Note that $\hat{c}_j$ cannot be computed from the data, so it must be ensured that the statistics we want to work with do not depend on the $\hat{c}_j$.
- ▶ To ensure the orthogonality of eigenfunctions, we must assume that $\lambda_1 > \lambda_2 > \dots$.
- ▶ In practice, we can estimate only the p largest eigenvalues, and assume that $\lambda_1 > \lambda_2 > \dots > \lambda_p > \lambda_{p+1}$, which implies that the first p eigenvalues are non-zero.

- With these preliminaries in mind, we define the estimated functional principal components (EFPC's) by

$$\int_0^1 \hat{\mathcal{K}}(t, s) \hat{v}_j(s) ds = \hat{\lambda}_j \hat{v}_j(t), \quad j = 1, 2, \dots, p; \quad t \in \mathcal{I}.$$

Theorem

Suppose $E\|X\|^4 < \infty$. Then, for each $1 \leq j \leq p$,

$$\limsup_{n \rightarrow \infty} nE(\|\hat{v}_j - v_j\|^2) < \infty, \quad \limsup_{n \rightarrow \infty} nE(|\lambda_j - \hat{\lambda}_j|^2) < \infty.$$

Testing hypothesis for model parameters

- ▶ Suppose we know that the parameter has to lie in one of two sets: either in Θ_0 or in Θ_1 , where $\Theta_0 \cap \Theta_1 = \emptyset$.
- ▶ We wish to employ the sample $X_1, \dots, X_n$ that we have at our disposal to decide which is the case.
- ▶ So we have the *null hypothesis*:

$$H_0 : \theta \in \Theta_0$$

and the competing *alternative hypothesis*:

$$H_1 : \theta \in \Theta_1.$$

- ▶ The statistical problem of hypothesis testing considers how to efficiently employ the sample to decide between two competing hypotheses H_0 and H_1 .

We assume the sample space is $\mathbb{S}$.

Definition [of test function]

A test function δ is any function $\delta : \mathbb{S}^n \rightarrow \{0, 1\}$.

The test function takes the value '0' when we rule in favour of H_0 based on the sample and takes the value '1' when we rule in favour of H_1 .

- ▶ Test functions are usually constructed by

$$\delta(X_1, \dots, X_n) = \begin{cases} 1, & \text{if } T(X_1, \dots, X_n) \in \mathcal{C}, \\ 0, & \text{if } T(X_1, \dots, X_n) \notin \mathcal{C}. \end{cases}$$

- ▶ T is called a *test statistic* and $\mathcal{C}$ a set in the range of T is called the *critical region*.
- ▶ In compact notation, we may write

$$\delta(x_1, \dots, x_n) = \mathbf{1}_{\mathcal{C}}\{T(x_1, \dots, x_n)\}.$$

- ▶ How to choose T and $\mathcal{C}$ in order to obtain a good test function?

Definition [error probabilities]

Let $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \in \Theta_1$ be two competing hypothesis.

- ▶ The Type I error Probability is defined to be the function

$$h(\theta) = \mathbb{P}_\theta(\delta(X_1, \dots, X_n) = 1), \quad \theta \in \Theta_0.$$

- ▶ The Type II error Probability is defined to be the function

$$g(\theta) = \mathbb{P}_\theta(\delta(X_1, \dots, X_n) = 0), \quad \theta \in \Theta_1.$$

$\mathbb{P}_\theta$ indicates that the random sample $X_1, \dots, X_n$ corresponds to the parameter θ .

Definition [Neyman-Pearson framework]

Let $H_0 : \theta \in \Theta_0$ and $H_1 : \theta \in \Theta_1$ be two competing hypothesis.

1. Fix an $\alpha \in (0, 1)$ and call it the *significance level* of the test.
2. Consider only test functions $\delta : \mathbb{S}^n \rightarrow \{0, 1\}$ such that

$$\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\delta = 1) = \alpha.$$

Call this class $\mathcal{D}(\Theta_0, \alpha)$.

3. Within the class of test functions $\mathcal{D}(\Theta_0, \alpha)$ compare test functions by considering which has higher power

$$\beta(\theta) = \mathbb{P}_\theta(\delta = 1), \quad \theta \in \Theta_1.$$

Definition [Optimal test]

A test function δ of $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$ is called optimal at level α if the following hold:

1. $\delta \in \mathcal{D}(\Theta_0, \alpha)$;
2. $\mathbb{P}_{\theta_1}(\phi = 1) \leq \mathbb{P}_{\theta_1}(\delta = 1)$ for all $\theta_1 \in \Theta_1$ and all $\phi \in \mathcal{D}(\Theta_0, \alpha)$.

One-Sample Problems for Functional Data

- We have a random functional sample

$$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Sp}(\mu, \gamma)$$

and its generic element is $X = (X(t), t \in \mathcal{I})$ ($X \in L_2(\mathcal{I})$).

- We wish to test the following hypothesis:

$$H_0 : \quad \mu(t) = \mu_0(t), \quad t \in \mathcal{I},$$

versus either

$$H_1 : \quad \mu(t) \neq \mu_0(t), \quad \text{for some } t \in \mathcal{I},$$

or

$$H_1 : \quad \|\mu - \mu_0\|^2 = \int_0^1 (\mu(t) - \mu_0(t))^2 dt \neq 0,$$

where $\mu_0(t)$ is some known function that is pre-specified based on related physical theories, past experiences, or past experimental results.

- We, first of all, have the following simple result.

Theorem

If $X_1, \dots, X_n \sim GSp(\mu, \gamma)$, then

$$\Delta(t) := \sqrt{n}(\hat{\mu}(t) - \mu(t)) \sim \mathcal{N}(0, \gamma(t, t))$$

and

$$(n-1)\hat{\gamma}(s, t) \sim WSp(n-1, \gamma).$$

- This theorem indicates that under the Gaussian assumption, the sample mean function is a Gaussian process, and a sample covariance function is proportional to a Wishart process.
- This result can be used for constructing various tests for the one-sample problem when the Gaussian assumption is satisfied.

Wishart process

- ▶ Wishart processes are natural generalizations of Wishart random matrices.
- ▶ Throughout, we use $WSp(n, \gamma)$ to denote a Wishart process with n degrees of freedom and a covariance function $\gamma(s, t)$.
- ▶ A general Wishart process $W(s, t) \sim WSp(n, \gamma)$ can be written as

$$W(s, t) = \sum_{i=1}^n W_i(s, t) = \sum_{i=1}^n v_i(s)v_i(t),$$

where $W_i(s, t) = v_i(s)v_i(t)$, $i = 1, 2, \dots, n$, i.i.d. $WSp(1, \gamma)$ and

$$v_i(t), i = 1, 2, \dots, n, \sim GSp(0, \gamma).$$

- ▶ The above theorem indicates that under the Gaussian assumption, the sample mean function is a Gaussian process, and a sample covariance function is proportional to a Wishart process.
- ▶ This result will be used for constructing various tests for the one-sample problem when the Gaussian assumption is satisfied.
- ▶ When the functional sample is not Gaussian but the sample size n is large, we use the following theorem.

Theorem

If the functional sample is with $\text{tr}(\gamma) < \infty$, then as $n \rightarrow \infty$, we have

$$\sqrt{n}(\hat{\mu} - \mu) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} Z \sim GSp(0, \gamma), \text{ in } L_2(\mathcal{I}), \quad (\text{CinD})$$

where $\xrightarrow[n \rightarrow \infty]{\mathcal{D}}$ denotes convergence in distribution.

$Z_n \xrightarrow[n \rightarrow \infty]{\mathcal{D}} Z$ in $L_2(\mathcal{I})$ means $\lim_{n \rightarrow \infty} EF(Z_n) = EF(Z)$ for each bounded continuous function $F : L_2(\mathcal{I}) \rightarrow \mathbb{R}$.

- Particularly, the convergence (CinD) gives for each continuous function $F : L_2(\mathcal{I}) \rightarrow \mathbb{R}$,

$$\lim_{N \rightarrow \infty} P(F(\Delta_n - \mu_\Delta) \leq x) = P(F(Z) \leq x)$$

for each x where the function $P(F(Z) \leq x)$ is continuous.

- Examples of continuous functions $F : L_2(\mathcal{I}) \rightarrow \mathbb{R}$ include

$$F(f) = \int_0^1 f^2(t) dt,$$

$$F(f) = \int_0^1 f(t)g(t)dt, \quad g \in L_2(\mathcal{I}).$$

To test the null hypothesis H_0 set

$$\Delta(t) = \sqrt{n}[\hat{\mu}(t) - \mu(t)]$$

as the pivotal test function.

Pointwise Tests

- Based on the point-wise estimators, the pivotal test statistic for the above local hypothesis testing problem is

$$Z_n(t) = \frac{\Delta(t)}{\sqrt{\hat{\gamma}(t, t)}} = \frac{\sqrt{n}[\hat{\mu}(t) - \mu_0(t)]}{\sqrt{\hat{\gamma}(t, t)}}.$$

- When the functional sample is Gaussian, then, for each $t \in \mathcal{T}$, we have

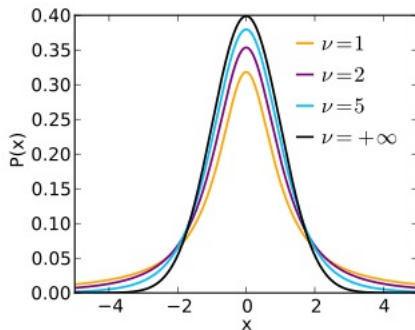
$$Z_n(t) \sim t_{n-1},$$

where t_{n-1} has Student's t -distribution with the parameter $n - 1$.

- Recall Student's t-distribution has the probability density function given by

$$f(t) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}},$$

where ν is the number of degrees of freedom and Γ is the gamma function, $\Gamma(s) = \int_0^\infty u^{s-1} e^{-u} du$.



- The point-wise t -test is conducted by rejecting H_{0t} whenever

$$|Z_n(t)| > t_{\alpha/2}(n-1),$$

where $t_{\alpha/2}(n-1)$ denotes the $100(1 - \alpha/2)$ percentile or $\alpha/2$ critical value of the t distribution with $n - 1$ degrees of freedom.

- Alternatively, we can construct the $100(1 - \alpha)\%$ point-wise confidence bands as

$$\hat{\mu}(t) \pm t_{\alpha/2}(n-1) \sqrt{\hat{\gamma}(t, t)/n}, \quad t \in \mathcal{I}.$$

- ▶ When the functional sample is not Gaussian, and n is small, then a nonparametric bootstrap approach can be used to bootstrap the critical values.

- ▶ Let

$$X_i^*(t), \quad i = 1, 2, \dots, n,$$

be a bootstrap sample randomly generated from (X_i) .

- ▶ Then, we can construct the sample mean and covariance functions as $\hat{\mu}^*(t)$ and $\hat{\gamma}^*(s, t)$, based on the bootstrap sample.
- ▶ The bootstrap test statistic $Z_n^*(t)$ is calculated by replacing $\hat{\mu}(t)$, $\hat{\gamma}(t, t)$, and $\mu_0(t)$ by $\hat{\mu}^*(t)$, $\hat{\gamma}^*(t, t)$, and $\hat{\mu}(t)$, respectively.
- ▶ This is because the bootstrap sample has the known mean function $\hat{\mu}(t)$.
- ▶ Repeat the above bootstrapping process many times, calculate its $100(1 - \alpha/2)$ -percentile, and then conduct the pointwise test or construct the pointwise bootstrap confidence bands accordingly.

L_2 -norm-based test

- ▶ The L_2 -norm-based test uses the squared L_2 -norm of $\Delta(t)$ as its test statistic:

$$T_n = \int_0^1 \Delta^2(t) dt = n \int_0^1 [\hat{\mu}(t) - \mu_0(t)]^2 dt.$$

- ▶ We see that T_n will be small under the null hypothesis $H_0 : \mu = \mu_0$ and it will be large under the alternatives $H_1 : \|\mu - \mu_0\|^2 = \int_0^1 (\mu(t) - \mu_0(t))^2 dt \neq 0$.

Theorem

Under the hypothesis that $X \sim \text{GSp}(\mu_0, \gamma)$, for any $a \geq 0$,

$$\mathbb{P}(T_n > a) = \mathbb{P}\left(\sum_{k=1}^{\infty} \lambda_k \gamma_k^2 > a\right),$$

where (η_k) standard normal random variables and (λ_k) are eigenvalues of the covariance γ .

- If X is not Gaussian but $\text{tr}(\gamma) < \infty$, then for large n ,

$$\mathbb{P}(T_n > a) \approx \mathbb{P}\left(\sum_{k=1}^{\infty} \lambda_k \gamma_k^2 > a\right).$$

- By the CLT,

$$\mathbb{P}\left(\left|\left(\sum_{k=1}^m \lambda_k \gamma_k^2 - \sum_{k=1}^m \lambda_k\right)\right| > a \left(\sum_{k=1}^m \lambda_k^2\right)^{1/2}\right) \approx \mathbb{P}(|Y| > a),$$

for large m , where $Y \sim \mathcal{N}(0, 1)$.

F-type test

- If γ is unknown, then it is natural to test H_0 using the following test statistic:

$$F_n = \frac{\int_0^1 \Delta^2(t) dt}{\text{tr}(\hat{\gamma})} = \frac{n \int_0^1 (\hat{\mu}(t) - \mu_0(t))^2 dt}{\text{tr}(\hat{\gamma})}.$$

Bootstrap test

- ▶ When the sample is not Gaussian and n is small, the null distributions of the L_2 -norm-based test is generally not tractable.
- ▶ In this case, we may use some nonparametric bootstrap approach to bootstrap the critical values of the L_2 -norm-based test resulting in the L_2 -norm-based bootstrap test.
- ▶ Let

$$X_i^*(t), \quad i = 1, 2, \dots, n$$

be a bootstrap sample randomly generated from the sample $x_1, \dots, x_n$.

- ▶ We construct the sample mean and sample covariance functions as $\hat{\mu}^*(t)$ and $\hat{\gamma}^*(t)$

- The L_2 -norm-based bootstrap test computes the bootstrap test statistic

$$T_n^* = \int_{\mathcal{I}} (\Delta^*(t))^2 dt$$

with the bootstrapped pivotal test function

$$\Delta^*(t) = \sqrt{n}[\hat{\mu}^*(t) - \hat{\mu}(t)].$$

- Repeat this process many times to obtain a bootstrap sample of T_n^* that can be used to estimate the $100(1 - \alpha)$ -percentile of T_n .

Remark

The bootstrap tests are easy to understand and implement, but they are time-consuming. They can be used when the sample size is small.

Two sample problem

- ▶ A general two-sample problem for functional data with a common covariance function can be formulated as follows.
- ▶ Suppose we have two functional samples

$$X_1(t), \dots, X_n(t) \stackrel{iid}{\sim} Sp(\mu_1, \gamma)$$

and

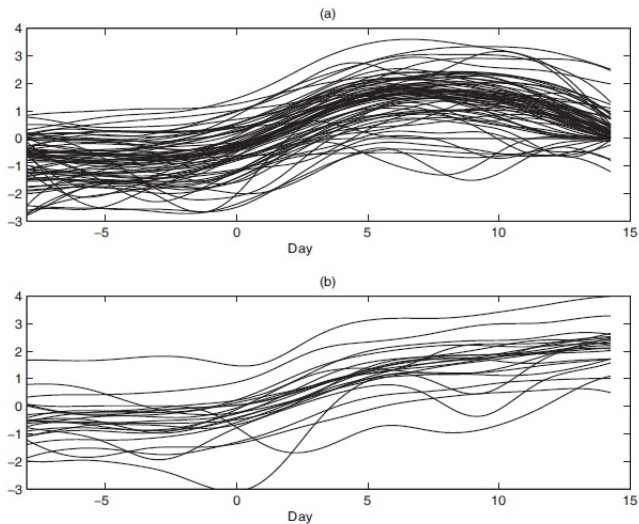
$$Y_1(t), \dots, Y_m(t) \stackrel{iid}{\sim} Sp(\mu_2, \gamma),$$

- ▶ where $\mu_1(t)$ and $\mu_2(t)$ are the unknown mean functions of the two samples, and $\gamma(s, t)$ is their common covariance function, which is usually unknown.
- ▶ We wish to test the following hypotheses:

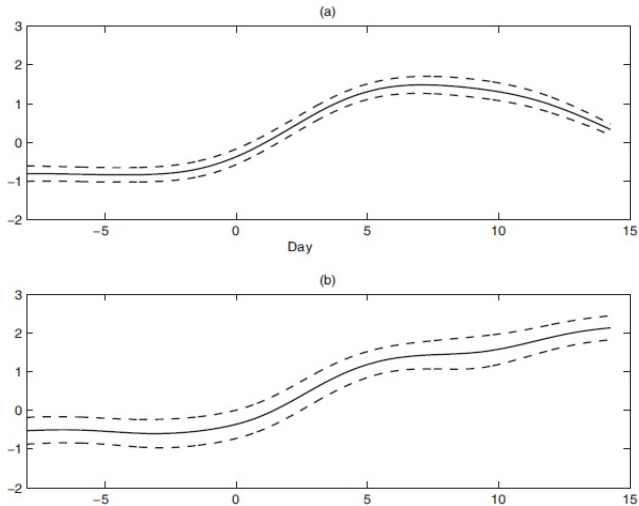
$$H_0 : \mu_1(t) = \mu_2(t), \quad t \in \mathcal{T}, \quad \text{versus}$$

$$H_1 : \mu_1(t) \neq \mu_2(t), \quad \text{for some } t \in \mathcal{T},$$

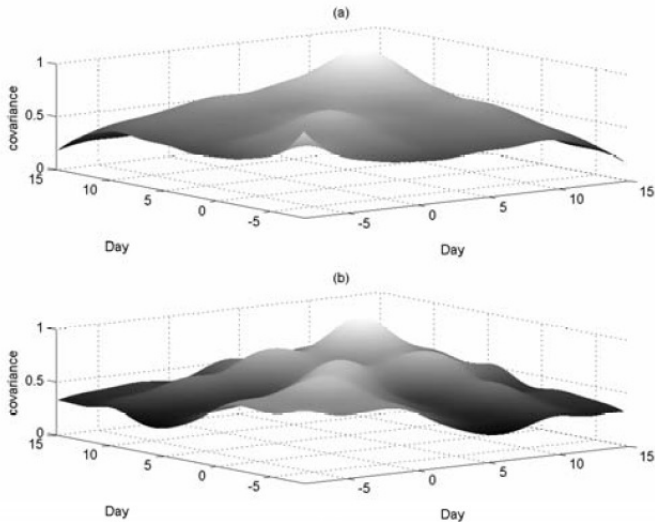
where $\mathcal{T}$ is the time period of interest, often a finite interval $[a, b] \subset \mathcal{I}$ say.



Reconstructed individual curves of the progesterone data: (a) and (b) samples, obtained by the local linear reconstruction method.



Sample mean functions (solid) of the progesterone data: (a) and (b) samples, together with their 95% pointwise confidence bands (dashed).



Sample covariance functions of the progesterone data: (a) and (b) samples. It appears that the two sample covariance functions are similar in shape.

- Based on the two functional samples, the unbiased estimators of the mean functions $\mu_1(t)$, $\mu_2(t)$ and the common covariance function $\gamma(s, t)$ are given by

$$\hat{\mu}_1(t) = n^{-1} \sum_{i=1}^n X_i(t), \quad \hat{\mu}_2(t) = m^{-1} \sum_{i=1}^m Y_i(t),$$

and

$$\hat{\gamma}(s, t) = (N - 1)^{-1} \left[\sum_{i=1}^n [X_i(s) - \hat{\mu}_1(s)][X_i(t) - \hat{\mu}_1(t)] + \sum_{j=1}^m [Y_j(s) - \hat{\mu}_2(s)][Y_j(t) - \hat{\mu}_2(t)] \right],$$

- which are known as the sample mean and pooled sample covariance functions of the two samples, respectively, where $N = n + m$ denotes the total sample size of the two samples.

- To test the two-sample problem based on the two samples, a natural the pivotal test function is

$$\Delta(t) = \sqrt{nm/N}[\hat{\mu}_1(t) - \hat{\mu}_2(t)]$$

which is the scaled mean function difference between the two samples.

- When the null hypothesis is valid, this quantity will be small and large otherwise.
- Therefore, it is appropriate to use $\Delta(t)$ as a pivotal test function for the two sample problems.
- Notice that $\Delta(t)$ has its mean and covariance functions as

$$\mu_{\Delta} := E\Delta(t) = \sqrt{nm/N}[\mu_1(t) - \mu_2(t)]$$

and

$$\text{cov}[\Delta(s), \Delta(t)] = \gamma(s, t).$$

- Under the null hypothesis we have $E\Delta(t) = 0, t \in T$.

- Before we can study any procedures for testing two sample hypotheses, we need to study the properties of $\Delta(t)$.

Theorem

Under Assumptions

- (i) The two samples (X_i) and (Y_i) are Gaussian;
 - (ii) $\mu_1(t), \mu_2(t) \in L_2(\mathcal{I})$ and $\text{tr}(\gamma) < \infty$,
- we have

$$\Delta(t) \sim GSp(\mu_\Delta, \gamma), \quad \text{and} \quad (n-2)\hat{\gamma}(s, t) \sim WSp(n-2, \gamma).$$

- Based on this, we can easily construct various tests for two sample problems.
- Theorem holds for finite functional samples but requires that the Gaussian assumption be satisfied. In practice, this Gaussian assumption is not always valid.

- In many cases, we can show that $\Delta(t)$ is asymptotically a Gaussian process.

Theorem

Under Assumptions

- (i) $\mu_1(t), \mu_2(t) \in L_2(\mathcal{I})$ and $\text{tr}(\gamma) < \infty$;
- (ii) as $N \rightarrow \infty$, the sample sizes satisfy $n/N \rightarrow \tau \in (0, 1)$;
- (iii) the processes $(X_i - \mu_1), (Y_i - \mu_2)$ are independent and identically distributed,

we have

$$\Delta(t) - \mu_{\Delta}(t) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} Z(t) \sim GSp(0, \gamma) \text{ in } L_2(\mathcal{I}). \quad (\text{CinD})$$

- The above theorem can be used for constructing various tests for two sample problem when the two samples are not Gaussian but with large sample sizes.

Theorem

Under some Assumptions, as $n \rightarrow \infty$, we have

$$\sqrt{n}(\hat{\gamma}(s, t) - \gamma(s, t)) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} GSp(0, \eta),$$

where

$$\eta((s_1, t_1), (s_2, t_2)) = E[v_{11}(s_1)v_{11}(t_1)v_{11}(s_2)v_{11}(t_2)] - \gamma(s_1, t_1)\gamma(s_2, t_2).$$

Point-wise Tests

- ▶ The key idea of a the point-wise test is to test the null hypothesis at each time point $t \in \mathcal{T}$.
- ▶ For any fixed $t \in \mathcal{T}$, the sub-problem is

$$H_{0t} : \mu_1(t) = \mu_2(t), \quad \text{versus} \\ H_{1t} : \mu_1(t) \neq \mu_2(t).$$

- Based on the sample mean functions and the pooled sample covariance function, the pivotal test statistic is

$$Z(t) = \frac{[\hat{\mu}_1(t) - \hat{\mu}_2(t)]}{\sqrt{(1/n + 1/m)\hat{\gamma}(t, t)}} = \frac{\Delta(t)}{\sqrt{\hat{\gamma}(t, t)}}.$$

- In many situations, the two samples may be approximately Gaussian, that is, under H_{0t} , we have

$$Z(t) \sim t(n-2), \quad t \in \mathcal{T}.$$

- Then, we can conduct the point-wise t -test by rejecting H_{0t} whenever

$$|Z(t)| > t_{\alpha/2}(n-2)$$

or by reporting the point-wise P -values computed based on the t distribution.

- ▶ When the Gaussian assumption is not satisfied for large samples, one may use the pointwise Z -test instead.
- ▶ As $n, m \rightarrow \infty$, we have

$$Z(t) \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \mathcal{N}(0, 1),$$

for any fixed $t \in \mathcal{T}$.

- ▶ Thus, one can conduct the point-wise Z -test by rejecting H_{0t} whenever

$$|Z(t)| > z_{\alpha/2}$$

or computing the point-wise P -values.

- ▶ When the two samples are not Gaussian while both n and m are small, the above point-wise t - and Z -tests are not preferred.
- ▶ In this case, one may resort to a point-wise bootstrap test.

L_2 -Norm-Based Test

- For the two-sample problem, the L_2 -norm-based test uses the squared L_2 -norm of the pivotal test function $\Delta(t)$ as the test statistic:

$$T_N = \int_0^1 \Delta^2(t) dt = \frac{nm}{N} \int_0^1 [\hat{\mu}_1(t) - \hat{\mu}_2(t)]^2 dt.$$

- we have under some conditions either

$$\Delta(t) \sim GSp(0, \gamma)$$

or

$$\Delta(t) \sim GSp(0, \gamma) \text{ asymptotically.}$$

- This gives approximately

$$T_N \stackrel{\mathcal{D}}{=} \sum_{r=1}^{\infty} \lambda_r \chi_r^2, \quad (\chi_r) \stackrel{iid}{\sim} \mathcal{N}(0, 1),$$

where $\lambda_1, \lambda_2, \dots$ are all the positive eigenvalues of the common covariance function $\gamma(s, t)$.

Heteroscedastic two-sample problem

- Suppose we have two independent functional samples

$$X_1(t), \dots, X_n(t) \stackrel{iid}{\sim} Sp(\mu_1, \gamma_1)$$

and

$$Y_1(t), \dots, Y_m(t) \stackrel{iid}{\sim} Sp(\mu_2, \gamma_2),$$

- where $\mu_1(t)$ and $\mu_2(t)$ are the unknown mean functions, and $\gamma_1(s, t), \gamma_2(s, t)$ are their covariance functions, which are also usually unknown.
- We wish to test the following hypotheses:

$$H_0 : \mu_1(t) = \mu_2(t), \quad t \in \mathcal{T}, \quad \text{versus}$$

$$H_1 : \mu_1(t) \neq \mu_2(t), \quad \text{for some } t \in \mathcal{T},$$

where $\mathcal{T}$ is the time period of interest, often a finite interval $[a, b] \subset \mathcal{I}$ say.

Theorem

If $E\|X\|^4 < \infty$, $E\|Y\|^4 < \infty$, and

$$\frac{n}{n+m} \rightarrow \theta \in (0, 1),$$

then

$$T_N \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \int_0^1 \Gamma^2(t) dt,$$

where $\Gamma = (\Gamma(t), 0 \leq t \leq 1)$ is a Gaussian process satisfying $E\Gamma(t) = 0$ and

$$E[\Gamma(t)\Gamma(s)] = (1 - \theta)\gamma_1(t, s) + \theta\gamma_2(t, s), \quad t, s \in [0, 1].$$

- The distribution of the limit in depends on the unknown covariance functions γ_1 and γ_2 . According to the Karhunen-Loève expansion, we can assume that

$$\Gamma(t) = \sum_{k=1}^{\infty} \tau_k^{1/2} N_k \psi_k(t),$$

where the N_k are independent standard normal random variables, $\tau_1 > \tau_2 > \dots$ and $\psi_1, \psi_2, \dots$ are the eigenvalues and eigenfunctions of the operator determined by $(1 - \theta)\gamma_1 + \theta\gamma_2$.

- Clearly,

$$\int_0^1 \Gamma^2(t) dt = \sum_{k=1}^{\infty} \tau_k N_k^2.$$

So to provide a reasonable approximation for $\int_0^1 \Gamma^2(t) dt$, we only need to estimate the τ_k 's.

- This can be done easily using $\hat{\tau}_k$'s, the eigenvalues of the empirical covariance function

$$\begin{aligned}\hat{\gamma}^*(t, s) = & \frac{m}{m+n} \frac{1}{n} \sum_{i=1}^n (X_i(t) - \bar{X}_n(t))(X_i(s) - \bar{X}_n(s)) \\ & + \frac{n}{m+n} \frac{1}{m} \sum_{i=1}^m (Y_i(t) - \bar{Y}_n(t))(Y_i(s) - \bar{Y}_n(s)).\end{aligned}$$

- $\sum_{k=1}^d \hat{\tau}_k N_k^2$ provides an approximation to the limit in if d is large enough.

Theorem

Suppose, that

$$\int_0^1 (\mu_1(t) - \mu_2(t))^2 dt > 0;$$

then

$$T_N \xrightarrow[n \rightarrow \infty]{P} \infty.$$

Projection method

- ▶ Now we use projections onto the space determined by the leading eigenfunctions of the operator $\gamma = (1 - \theta)\gamma_1 + \theta\gamma_2$.
- ▶ We assume that the eigenvalues satisfy

$$\tau_1 > \tau_2 > \cdots > \tau_d > \tau_{d+1}.$$

- ▶ The corresponding eigenfunctions are $\phi_1, \dots, \phi_{d+1}$.
- ▶ Since these functional are unknown we use $\hat{\phi}_1, \dots, \hat{\phi}_{d+1}$.
- ▶ We project $\overline{X}_n - \overline{Y}_m$ into linear space spanned by $\hat{\phi}_1, \dots, \hat{\phi}_d$.
Let

$$\hat{\xi}_i = \langle \overline{X}_n - \overline{Y}_m, \hat{\phi}_i \rangle, \quad i = 1, \dots, d.$$

- ▶ Testing procedures can be based on the statistics

$$T_N^{(1)} = \frac{nm}{n+m} \sum_{k=1}^d \hat{\xi}_k^2 / \hat{\tau}_k,$$

- ▶ and

$$T_N^{(2)} = \frac{nm}{n+m} \sum_{k=1}^d \hat{\xi}_k^2.$$

Theorem

Under H_0 and if certain conditions are satisfied, then

$$T_N^{(1)} \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \chi^2(d)$$

and

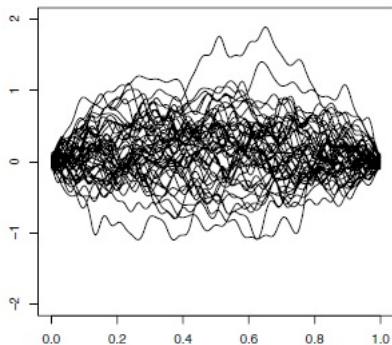
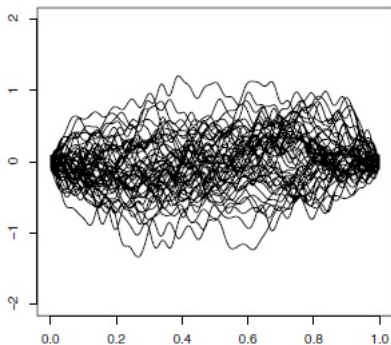
$$T_N^{(2)} \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \sum_{k=1}^d \tau_k N_k^2,$$

where $\chi^2(d)$ is a chi-square random variable with d degrees of freedom, and $N_1, \dots, N_d$ are independent standard normal random variables.

A simulation study

We compare the statistics $T_N^{(1)}$ and $T_N^{(2)}$ (the study is taken from Horvath and Kokoszka, Inference for functional data with applications).

- ▶ Consider sample sizes $n = 50$ and $n = 100$ and $m = n$ as well as $M = 2N$.
- ▶ To compare the sizes, we set $\mu_1(t) = \mu_2(t) = 0$.
- ▶ Under the alternative, we set $\mu_1(t) = 0$ and $\mu_2(t) = at(1 - t)$.
- ▶ The power is then a function of the parameter a .
- ▶ We consider two settings for the processes:
 - 1) Both the X_i and Y_i are Brownian bridges with corresponding shift.
 - 2) The X_i are Brownian bridges, and the Y_i are Brownian motions with drift.
- ▶ To compute the test statistics, we converted the Gaussian processes simulated as increments into functional objects using the 49 Fourier basis function.
- ▶ We then computed the test statistics with $d = 5$.



Fifty trajectories of the Brownian bridge (left) and fifty independent trajectories of the Brownian bridge plus $\mu_2(t) = 0.8t(1 - t)$ (right). The tests can detect the different means with a probability higher than 90%.

- ▶ Except one function in the right panel, which goes visibly above the other functions, both sets look very similar, and it would be difficult to tell by eye that they have different mean functions.
- ▶ Based on one thousand replications, $T_{50+50}^{(1)}$ rejects the null hypothesis of equal means with probability 0.91 and $T_{50+50}^{(2)}$ with probability 0.98, at the nominal size $\alpha = 0.05$.
- ▶ For such relatively small sizes, the tests suffer from an elevated probability of type I error. For $\alpha = 0.05$, the empirical size is 0.66 for $T_{50+50}^{(1)}$ and 0.73 for $T_{50+50}^{(2)}$.

a	$\alpha = .01$		$\alpha = .05$		$\alpha = .10$	
	$T_{100,200}^{(1)}$	$T_{100,200}^{(2)}$	$T_{100,200}^{(1)}$	$T_{100,200}^{(2)}$	$T_{100,200}^{(1)}$	$T_{100,200}^{(2)}$
0.0	1.3	1.4	5.6	5.7	10.8	10.8
0.1	1.7	1.7	6.8	6.4	12.2	12.1
0.2	2.6	3.0	9.8	11.0	16.5	18.2
0.3	4.8	6.1	15.8	18.5	24.7	27.3
0.4	9.8	10.7	23.6	27.7	35.6	39.0
0.5	18.5	19.5	37.8	41.5	50.1	54.6
0.6	29.3	29.7	52.3	55.0	64.3	67.2
0.7	42.7	43.1	65.5	67.5	75.6	78.1
0.8	59.2	57.7	79.3	80.3	87.1	88.1
0.9	73.7	71.0	88.7	89.0	93.7	94.7
1.0	85.3	82.0	94.4	94.8	97.2	97.7
1.1	92.8	89.8	97.8	97.3	98.9	98.7
1.2	96.6	94.7	99.3	99.2	99.9	99.7
1.3	98.9	97.7	99.8	99.7	99.9	99.9
1.4	99.5	99.3	100.0	99.9	100.0	100.0
1.5	99.8	99.8	100.0	100.0	100.0	100.0
1.6	100.0	100.0	100.0	100.0	100.0	100.0

Size ($a = 0$) and power ($a > 0$) of the tests based on $T_{100+200}^{(1)}$ and $T_{100+200}^{(2)}$. The sample with $n = 100$ has Brownian bridge errors, and the one with $m = 200$ has Brownian motion errors.

One-Way ANOVA

- ▶ In classical linear models, one-way analysis of variance (ANOVA) is a technique used to compare means of three or more samples or groups.
- ▶ The samples or groups are grouped according to a categorical variable known as a factor.
- ▶ The number of different values taken by the factor is called the level of the factor.

- ▶ We can define the one-way ANOVA problem for functional data as follows.
- ▶ Suppose we have k independent functional samples:

$$X_{i1}(t), \dots, X_{in_i}(t) \sim \text{Sp}(\mu_i, \gamma) \quad i = 1, \dots, k,$$

where $\mu_1(t), \mu_2(t), \dots, \mu_k(t)$ are the unknown group mean functions of the k samples and $\gamma(s, t)$ is the common covariance function.

- ▶ We wish to test the following one-way ANOVA testing problem:

$$H_0 : \mu_1(t) = \mu_2(t) = \dots = \mu_k(t), \quad t \in \mathcal{T},$$

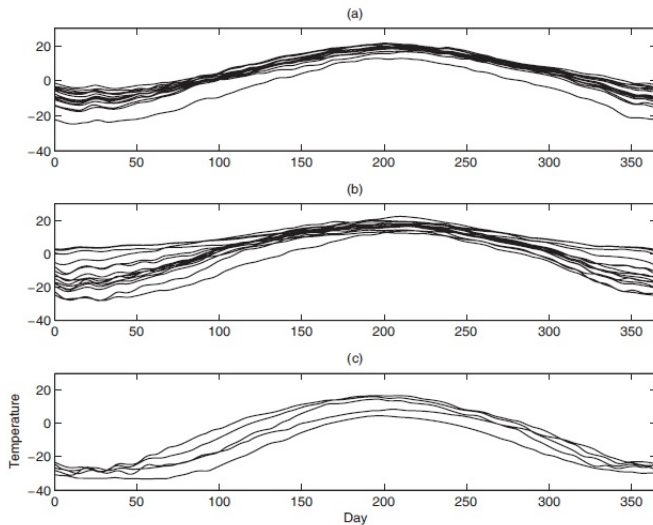
where again $\mathcal{T}$ is some time period of interest, often specified as $[a, b] \subseteq [0, 1]$.

- ▶ The above one-way ANOVA problem is also known as the k -sample problem for functional data, extending the two-sample problem for functional data.

- ▶ The one-way ANOVA testing problem often aims to check if the effect of a factor or a treatment is statistically significant.
- ▶ This factor or treatment is usually used to group the individual functions into several samples, groups or categories.
- ▶ If the factor or treatment seriously impacts the functional data, the one-way ANOVA problem will be statistically significant.

Example

In the Canadian temperature data, the factor is “the location of a weather station” that may affect the temperature of the weather station. The number, k , of samples or categories in which a functional data set is grouped is the number of levels of the factor. The number of levels for the Canadian temperature data is $k = 3$.



Reconstructed Canadian temperature curves for (a) fifteen Eastern weather stations, (b) fifteen Western weather stations, and (c) five Northern weather stations. The local linear kernel smoother with the Gaussian kernel is applied.

For the one-way ANOVA problem, one is interested mainly in the following three major kinds of tests:

Main-Effect Test

- Set

$$\mu_i(t) = \mu(t) + \alpha_i(t), \quad i = 1, 2, \dots, k,$$

where $\mu(t)$ is known as the overall mean function of the k samples and $\alpha_i(t)$ is the i 'th main-effect function for $i = 1, 2, \dots, k$.

- Then the model can be further written as the following standard one-way ANOVA model for functional data:

$$X_{ij}(t) = \mu(t) + \alpha_i(t) + \varepsilon_{ij}(t), \quad j = 1, 2, \dots, n_i; \quad i = 1, 2, \dots, k.$$

In this formulation, the null hypothesis can be equivalently expressed as

$$H_0(\alpha) : \quad \alpha_1(t) \equiv \alpha_2(t) \equiv \alpha_k(t) \equiv 0, \quad t \in \mathcal{I},$$

that is, to test if the main-effect functions are the same and are equal to 0.

Post Hoc Test

- ▶ When $H_0(\alpha)$ is accepted, the one-way ANOVA model is not significant. When it is rejected, further investigation is often required.
- ▶ For example, one may want to know if any two main-effect functions $\alpha_i(t)$ and $\alpha_j(t)$ are the same, where i and j are any two integers such that $1 \leq i < j \leq k$.
- ▶ This test can be written as

$$H_0 : \alpha_i(t) \equiv \alpha_j(t), \quad t \in \mathcal{I}$$

versus

$$H_1 : \alpha_i(t) \not\equiv \alpha_j(t), \quad \text{for some } t \in \mathcal{I}.$$

- ▶ The above test is known as a post hoc test.

Contrast Test

The post hoc tests are special cases of contrast tests.

- Let $a_1, \dots, a_k$ be k constants such that they add up to 0, that is,

$$\sum_{i=1}^k a_i = \mathbf{a}'\mathbf{1}_k = 0,$$

where $\mathbf{a} = [a_1, \dots, a_k]'$ and $\mathbf{1}_k$ is a column vector of k ones.

- A contrast is defined as

$$\sum_{i=1}^k a_i \alpha_i(t) = \mathbf{a}'\boldsymbol{\alpha}(t),$$

a linear combination of the main-effect functions $\alpha_i(t)$, $i = 1, 2, \dots, k$, where $\boldsymbol{\alpha}(t) = [\alpha_1(t), \dots, \alpha_k(t)]'$ consists of all the main-effect functions.

- ▶ A simple contrast is the difference of two main-effect functions, for example,

$$\alpha_1(t) - \alpha_2(t).$$

- ▶ Another simple example of contrast is

$$\alpha_1(t) - 3\alpha_2(t) + 2\alpha_5(t), \text{ when } k \geq 5.$$

- ▶ For a given $\mathbf{a} \in \mathbb{R}^k$ such that $\mathbf{a}'\mathbf{1}_k = 0$, a contrast test is defined as

$$H_0 : \mathbf{a}'\boldsymbol{\alpha}(t) \equiv 0, \quad t \in \mathcal{T}$$

versus

$$H_1 : \mathbf{a}'\boldsymbol{\alpha}(t) \neq 0, \text{ for some } t \in \mathcal{I}.$$

Estimation of Group Mean and Covariance Functions

- Based on the k samples, the group mean functions $\mu_i(t), i = 1, 2, \dots, k$ and the common covariance function $\gamma(s, t)$ can be unbiasedly estimated as

$$\hat{\mu}_i(t) = \overline{X_{i\cdot}}(t) = n_i^{-1} \sum_{j=1}^{n_i} X_{ij}(t), \quad i = 1, 2, \dots, k,$$

$$\hat{\gamma}(s, t) = (n - k)^{-1} \sum_{i=1}^k \sum_{j=1}^{n_i} [X_{ij}(s) - \hat{\mu}_i(s)][X_{ij}(t) - \hat{\mu}_i(t)],$$

where $n = \sum_{i=1}^k n_i$ denotes the total sample size.

- The estimated covariance function $\hat{\gamma}(s, t)$ is also known as the pooled sample covariance function.

- Note that $\hat{\mu}_i(t), i = 1, 2, \dots, k$ are independent and for $i = 1, 2, \dots, k$,

$$\begin{aligned}E(\hat{\mu}_i(t)) &= \mu_i(t), \\ \text{cov}(\hat{\mu}_i(s), \hat{\mu}_i(t)) &= \gamma(s, t)/n_i.\end{aligned}$$

- Set $\hat{\boldsymbol{\mu}}(t) = [\hat{\mu}_1(t), \hat{\mu}_2(t), \dots, \hat{\mu}_k(t)]'$. It is an unbiased estimator of $\boldsymbol{\mu}(t) = [\mu_1(t), \mu_2(t), \dots, \mu_k(t)]'$.
- We have $E\hat{\boldsymbol{\mu}}(t) = \boldsymbol{\mu}(t)$ and $\text{cov}(\hat{\boldsymbol{\mu}}(s), \hat{\boldsymbol{\mu}}(t)) = \gamma(s, t)D$, where $D = \text{diag}(1/n_1, 1/n_2, \dots, 1/n_k)$ is a diagonal matrix with diagonal entries $1/n_i, i = 1, 2, \dots, k$.
- Hence,

$$\hat{\boldsymbol{\mu}}(t) \sim \text{Sp}_k(\boldsymbol{\mu}, \gamma D),$$

where $\text{Sp}_k(\boldsymbol{\mu}, \Gamma)$ denotes a k -dimensional stochastic process having the vector of mean functions $\boldsymbol{\mu}(t)$ and the matrix of covariance functions $\Gamma(s, t)$.

Theorem

Under assumptions

- (i) The k samples are in $L_2(0, 1)$ and $\text{tr}(\gamma) < \infty$;
- (ii) The k samples are Gaussian,

we have

$$D^{-1/2}[\hat{\boldsymbol{\mu}}(t) - \boldsymbol{\mu}(t)] \sim \text{GSp}_k(0, \gamma \mathbf{I}_k), \quad \text{and} \\ (n - k)\hat{\gamma}(s, t) \sim \text{WSp}(n - k, \gamma).$$

- When the Gaussian assumption (ii) is not valid, we need more assumptions to derive the asymptotic distribution of $\hat{\mu}(t)$, including that the sample sizes $n_1, n_2, \dots, n_k$ should be proportionally large.

Theorem

Under Assumptions:

- (i) the k samples are in $L_2(0, 1)$ and $\text{tr}(\gamma) < \infty$;
 - (ii) as $n \rightarrow \infty$, the k sample sizes satisfy $n_i/n \rightarrow \tau_i, i = 1, 2, \dots, k$ with $\tau_1, \tau_2, \dots, \tau_k \in (0, 1)$;
 - (iii) the subject-effect functions $X_{ij}(t) - \mu_i(t), j = 1, 2, \dots, n_i; i = 1, 2, \dots, k$ are i.i.d.;
- as $n \rightarrow \infty$, we have

$$D^{-1/2}[\hat{\mu}(t) - \mu(t)] \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \text{GSp}_k(0, \gamma \mathbf{I}_k).$$

- With some further assumptions, we can show that the pooled sample covariance function $\hat{\gamma}(s, t)$ is asymptotically Gaussian.

- ▶ For the main-effect, post hoc, or contrast tests, we do not need to identify the main-effect functions $\alpha_i(t), i = 1, 2, \dots, k$.
- ▶ In fact, they are not identifiable unless some constraint is imposed.
- ▶ If we do want to estimate these main-effect functions, the most commonly used constraint is

$$\sum_{i=1}^k n_i \alpha_i(t) = 0,$$

involving the k sample sizes.

- Under this constraint, it is easy to show that the unbiased estimators of the main-effect functions are

$$\hat{\alpha}_i(t) = \bar{X}_{i.}(t) - \bar{X}_{..}(t), \quad i = 1, 2, \dots, k,$$

where

$$\bar{X}_{..}(t) = n^{-1} \sum_{i=1}^k \sum_{j=1}^{n_i} X_{ij}(t) = n^{-1} \sum_{i=1}^k n_i \bar{X}_{i.}(t)$$

is the usual sample *grand mean function*.

- Under the constraint, $\bar{X}_{..}(t)$ is an unbiased estimator of the grand mean function $\mu(t)$.

► Let

$$SSR_n(t) = \sum_{i=1}^k n_i [\bar{X}_{i.}(t) - \bar{X}_{..}(t)]^2,$$

and

$$SSE_n(t) = \sum_{i=1}^k \sum_{j=1}^{n_i} [X_{ij}(t) - \bar{X}_{i.}(t)]^2,$$

denote the pointwise between-subject and within-subject variations, respectively.

Theorem

Suppose Assumptions

- (i) the k samples are in $L_2(0, 1)$ and $\text{tr}(\gamma) < \infty$;
 - (ii) the k samples are Gaussian,
- hold. Then under the null hypothesis

$$H_0 : \mu_1(t) \equiv \mu_2(t) \equiv \cdots \equiv \mu_k(t), \quad t \in \mathcal{I},$$

we have

$$\int_{\mathcal{I}} SSR_n(t) dt \stackrel{\mathcal{D}}{=} \sum_{r=1}^m \lambda_r A_r, \quad A_r \stackrel{iid}{\sim} \chi_{k-1}^2,$$

and

$$\int_{\mathcal{I}} SSE_n(t) dt \stackrel{\mathcal{D}}{=} \sum_{r=1}^m \lambda_r E_r, \quad E_r \stackrel{iid}{\sim} \chi_{n-k}^2,$$

where $A_r, E_r, r = 1, 2, \dots, m$ are independent of each other, and $\lambda_1, \dots, \lambda_m$ are all the positive eigenvalues of $\gamma(s, t)$.

Theorem

Suppose the following assumptions hold,

- (i) the k samples are in $L_2(0, 1)$ and $\text{tr}(\gamma) < \infty$;
- (ii) as $n \rightarrow \infty$, the k sample sizes satisfy $n_i/n \rightarrow \tau_i, i = 1, 2, \dots, k$ with $\tau_1, \tau_2, \dots, \tau_k \in (0, 1)$;
- (iii) the subject t-effect functions $X_{ij}(t) - \mu_i(t), j = 1, 2, \dots, n_i; i = 1, 2, \dots, k$ are i.i.d..

Then under the null hypothesis

$$H_0 : \mu_1(t) \equiv \mu_2(t) \equiv \dots \equiv \mu_k(t), \quad t \in \mathcal{I},$$

as $n \rightarrow \infty$, we have

$$\int_{\mathcal{I}} SSR_n(t) dt \xrightarrow[n \rightarrow \infty]{\mathcal{D}} \sum_{r=1}^m \lambda_r A_r, \quad A_r \stackrel{iid}{\sim} \chi_{k-1}^2,$$

where $\lambda_1, \dots, \lambda_m$ are all the positive eigenvalues of $\gamma(s, t)$.

One-way ANOVA tests

- Let

$$X_{i1}(t), X_{i2}(t), \dots, X_{in_i}(t), \quad i = 1, \dots, k$$

denote k groups of random functions defined over a given finite interval $\mathcal{I} = [a, b]$.

- Let $\text{Sp}(\mu, \gamma)$ denote a stochastic process with mean function $\mu(t), t \in \mathcal{I}$ and covariance function $\gamma(s, t), s, t \in \mathcal{I}$.
- Assuming that $X_{i1}(t), X_{i2}(t), \dots, X_{in_i}(t)$ are i.i.d. $\text{Sp}(\mu_i, \gamma), i = 1, \dots, k$, it is often interesting to test the equality of the k mean functions:

$$H_0 : \mu_1(t) = \dots = \mu_k(t), \quad t \in \mathcal{I},$$

against the alternative that its negation holds.

- We first set up notation.
- Let

$$SSR_n(t) = \sum_{i=1}^k n_i (\bar{X}_{i.}(t) - \bar{X}_{..}(t))^2$$

and

$$SSE_n(t) = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij}(t) - \bar{X}_{i.}(t))^2$$

denote the point-wise between-subject and within-subject variations respectively, where

$$\bar{X}_{..}(t) = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} X_{ij}(t) \quad \text{and} \quad \bar{X}_{i.}(t) = \frac{1}{n_i} \sum_{j=1}^{n_i} X_{ij}(t),$$

$i = 1, \dots, k$, are respectively the sample grand mean function and the sample group mean functions.

- Let $\text{tr}(\gamma) = \int_{\mathcal{I}} \gamma(t, t) dt$ denote the trace of $\gamma(s, t)$.
- The pooled sample covariance function $\hat{\gamma}(s, t)$, as an unbiased estimator of $\gamma(s, t)$, is given by

$$\hat{\gamma}(s, t) = \frac{1}{n - k} \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij}(s) - \bar{X}_{i\cdot}(s))(X_{ij}(t) - \bar{X}_{i\cdot}(t)).$$

Point-wise F-test

- ▶ The test statistic of the point-wise F -test for H_0 is defined as

$$F_n(t) = \frac{SSR_n(t)/(k-1)}{SSE_n(t)/(n-k)}.$$

- ▶ The pointwise F -test rejects H_0 at each $t \in \mathcal{I}$ whenever

$$F_n(t) > F_\alpha(k-1, n-k)$$

for any predetermined significant level α , where

$F_\alpha(k-1, n-k)$ denotes the upper $100(1-\alpha)$ percentile of $F_{k-1, n-k}$.

A percentile (or a centile) is a measure used in statistics indicating the value below which a given percentage of observations in a group of observations falls.

- ▶ Hence we can test H_0 at all points of $\mathcal{I}$ using the same critical value for any predetermined significant level.
- ▶ Unfortunately, the point-wise F -test is time-consuming (it must be performed for all $t \in \mathcal{I}$) and it is not guaranteed that the one-way ANOVA problem is significant overall for a given significance level even when the point-wise F -test is significant for all $t \in \mathcal{I}$ at the same significance level.

- ▶ If a random variable Y has an F -distribution with parameters d and m , we write $Y \sim F_{d,m}$.
- ▶ If $U_1 \sim \chi_d^2$, $U_2 \sim \chi_m^2$ and are independent, then

$$Y = \frac{U_1/d}{U_2/m} \sim F_{d,m}.$$

- ▶ The probability density function (pdf) for $Y \sim F_{d,m}$ is given by

$$f_{d,m}(x) = \frac{1}{B(d/2, m/2)} \left(\frac{d}{m}\right)^{\frac{d}{m}} x^{\frac{d}{2}-1} \left(1 + \frac{d}{m} x\right)^{-\frac{d+m}{2}}$$

CH-test

- CH-test uses the following test statistic:

$$V_n = \sum_{1 \leq i < j \leq k} n_i \int_{\mathcal{I}} (\bar{X}_{i.}(t) - \bar{X}_{j.}(t))^2 dt.$$

- Under the null hypothesis H_0 and the assumptions that $n_i, n \rightarrow \infty$ in such a way that $n_i/n \rightarrow p_i > 0$ for $i = 1, \dots, k$, the approximate distribution of V_n is that of the statistic

$$V = \sum_{1 \leq i < j \leq k} n_i \int_{\mathcal{I}} (Z_i(t) - \sqrt{p_i/p_j} Z_j(t))^2 dt,$$

where $Z_1(t), \dots, Z_k(t)$ are independent Gaussian processes with mean 0 and covariance function $\gamma(s, t)$.

L_2 -Norm-Based Test

- ▶ Test statistic is defined as the integral of the point-wise between-subject variations:

$$T_n = \int_{\mathcal{I}} SSR_n(t) dt = \sum_{i=1}^k n_i \int_{\mathcal{I}} [\bar{X}_{i.}(t) - \bar{X}_{..}(t)]^2 dt.$$

- ▶ Under the null hypothesis and under certain conditions, we have approximately

$$T_n = \sum_{r=1}^m \lambda_r A_r^2, \quad (A_r) \stackrel{iid}{\sim} \chi_{k-1}^2,$$

where λ_r , $r = 1, 2, \dots, m$ are all the positive eigenvalues of $\gamma(s, t)$.

Globalized Point-wise F -tests

- ▶ GPF test, whose test statistic is taken as the integral of the point-wise F -test statistic over $\mathcal{I}$:

$$T_n = \int_{\mathcal{I}} F_n(t) dt.$$

- ▶ Via some simulation studies, it is shown that in terms of size-controlling and power, the GPF test is, in general, comparable with the L_2 -norm based test and the F -type test adopted for the one-way ANOVA problem.
- ▶ Alternatively, one can also globalize the point-wise F -test using the supremum of the point-wise F test statistic over $\mathcal{I}$:

$$F_{\max,n} = \sup_{t \in \mathcal{I}} F_n(t).$$

Condition A

- (1) The k population mean functions $\mu_1(t), \mu_2(t), \dots, \mu_k(t)$ all belong to $L_2(\mathcal{I})$.
- (3) As $n \rightarrow \infty$, the k sample sizes $n_1, \dots, n_k$ satisfy $n_i/n \rightarrow \tau_i$, $i = 1, 2, \dots, k$, such that $\tau_1, \tau_2, \dots, \tau_k \in (0, 1)$.
- (4) The subject-effect function $v_{11}(t)$ satisfies
$$E\|v_{11}\|^4 = E\left(\int_{\mathcal{I}} v_{11}^2(t) dt\right)^2 < \infty.$$
- (5) The covariance function $\gamma(s, t)$ satisfies $\gamma \in \mathcal{H}_{2\beta}(\mathcal{I} \times \mathcal{I})$ with $\text{tr}(\gamma) < \infty$, where $0 < 2\beta < 1$. For any $t \in \mathcal{I}$, $\gamma(t, t) > 0$.
- (6) The expectation $E[v_{11}^2(s)v_{11}^2(t)]$ is uniformly bounded. That is, for any $(s, t) \in \mathcal{I} \times \mathcal{I}$, we have $E[v_{11}^2(s)v_{11}^2(t)] \leq C < \infty$, where C is some constant independent of (s, t) .

Theorem

Under Condition A and the null hypothesis, as $n \rightarrow \infty$, we have

$$F_{\max, n} \xrightarrow[n \rightarrow \infty]{\mathcal{D}} R_0$$

with

$$R_0 \sim \sup_{t \in \mathcal{I}} \left\{ (k-1)^{-1} \sum_{i=1}^{k-1} w_i^2(t) \right\},$$

where

$$w_1, w_2, \dots, w_{k-1} \stackrel{iid}{\sim} GSp(0, \gamma_w)$$

with

$$\gamma_w(s, t) = \frac{\gamma(s, t)}{\sqrt{\gamma(t, t)\gamma(s, s)}}.$$

- The covariance function $\gamma_w(s, t)$ can be estimated as

$$\hat{\gamma}_w(s, t) = \frac{\hat{\gamma}(s, t)}{\sqrt{\hat{\gamma}(t, t)\hat{\gamma}(s, s)}},$$

where $\hat{\gamma}(s, t)$ is the pooled sample covariance function.

- We can sample the i.i.d. Gaussian processes $w_i, i = 1, 2, \dots, k - 1$, from $GSp(0, \hat{\gamma})$ a large number of times so that a large sample of R_0 can be obtained and hence the desired upper 100-percentile of R_0 can be computed accordingly.
- However this method approximates the critical values well only for large sample sizes.

- Denote the estimated subject-effect functions as

$$\hat{v}_{ij}(t) = X_{ij}(t) - \bar{X}_{i\cdot}(t), \quad j = 1, 2, \dots, n_i; \quad i = 1, 2, \dots, k,$$

which can be regarded as estimators of the subject-effect functions $v_{ij}(t), j = 1, 2, \dots, n_i; \quad i = 1, 2, \dots, k$.

- Let

$$v_{ij}^*(t), \quad j = 1, 2, \dots, n_i; \quad i = 1, 2, \dots, k,$$

be bootstrapped k samples randomly generated from the estimated subject-effect functions.

- The non-parametric bootstrapped $F_{max,n}$ test statistic can then be obtained as

$$F_{max,n}^* = \sup_{t \in \mathcal{I}} F_n^*(t), \quad \text{where} \quad F_n^*(t) = \frac{SSR_n^*(t)/(k-1)}{SSE_n^*(t)/(n-k)}.$$

- ▶ Repeat the above bootstrapping process a large number of times and calculate the upper 100-percentile of the bootstrap sample on $F_{max,n}^*$ and then conduct the α -level F_{max} -test using this as the critical value.
- ▶ Let C_α and C_α^* denote the upper 100-percentiles of R_0 and $T_{max,n}^*$ respectively.

Theorem

Under Condition A, as $n \rightarrow \infty$, we have

$$F_{max,n}^* \xrightarrow[n \rightarrow \infty]{\mathcal{D}} R_0 \quad \text{and} \quad C_\alpha^* \xrightarrow[n \rightarrow \infty]{\mathcal{D}} C_\alpha.$$

- ▶ Notice that Proposition holds under both the null and the alternative hypotheses. This is a desirable property since in practice either of the null and alternative hypotheses may be true. This property is mainly due to the fact that given the original k samples, the k bootstrapped samples all have the same group mean function as 0 and hence the null hypothesis always holds for the bootstrapped k samples.

Change point problem

- ▶ Change-point detection is a long-standing question in mathematical statistics, which has attracted much attention since the 30s.
- ▶ In the time series study, it is natural to assume stationarity, that is, time-shift invariance of the data probability distribution.
- ▶ Indeed, suppose that we are recording data coming from a natural phenomenon during a time period where there is no trend or shock in the background; then, there is no reason for the distribution parameters to change.
- ▶ This assumption often fails in practice, where the environment does endure potentially large changes and a more reasonable assumption is to consider that the observed phenomenon is stationary only on smaller time-units

- ▶ Interest in the statistical analysis of change point detection and estimation has attracted considerable attention in the fields of data mining and statistics mainly because change point problems can be encountered in many disciplines such as economics, finance, medicine, psychology, geology, computer science, literature, etc.
- ▶ From the statistical point of view, a **change point is a place or time point such that the observations follow one distribution up to that point and follow another distribution after that point.**

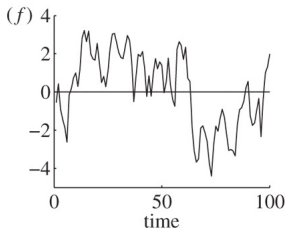
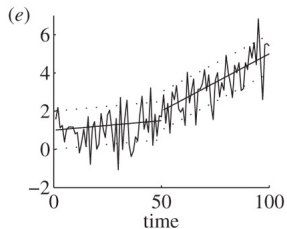
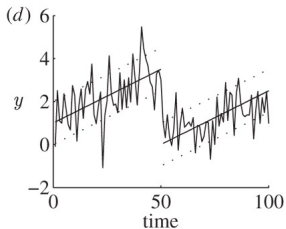
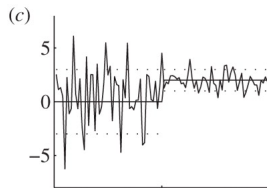
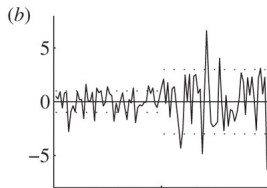
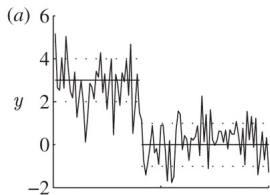
One change point

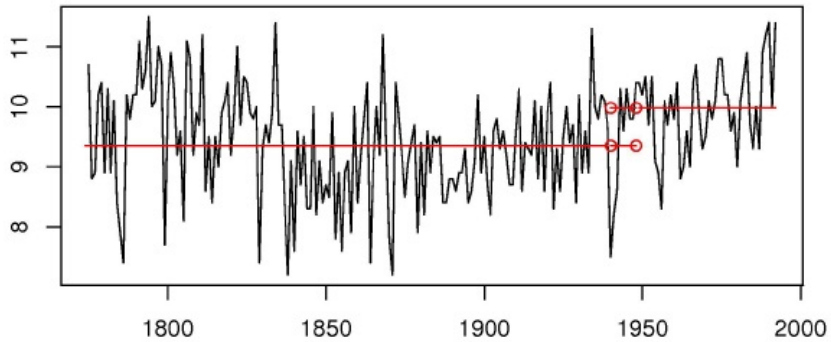
- (One) change point problem can be defined in the following way:

for the sequence $Y = \{Y_k, k = 1, \dots, n, \}$ there is $1 < k^* < n$ such that Y_j is following some distribution P for $1 \leq j \leq k^*$ and another distribution P^* for $j > k^*$.

- The point k^* is defined as a change point.

One change point





Multiple change point

- Multiple change point problems can be defined in the following way:

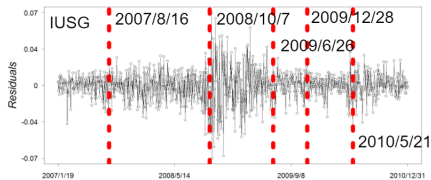
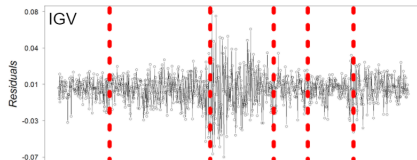
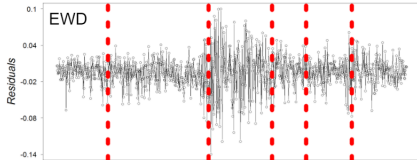
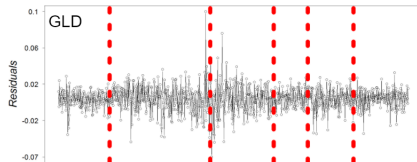
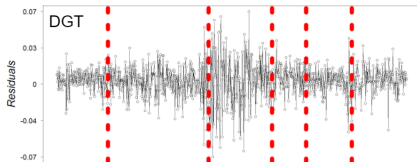
we can view a univariate process $Y = \{Y_t, t = 1, \dots, n\}$ as following some distribution P_i between the $(i - 1)$ 'th and the i 'th change points

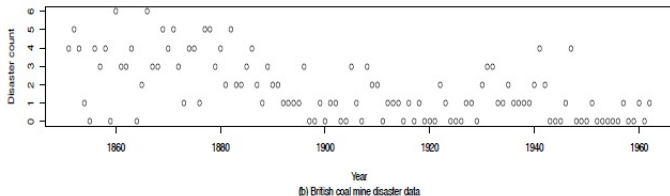
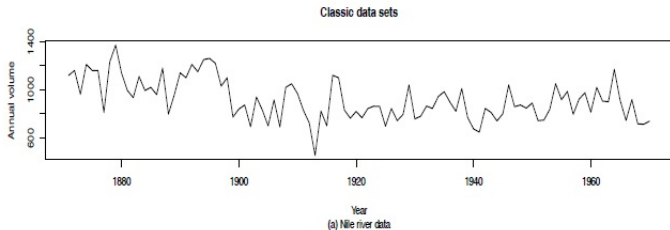
$$Y_t \sim P_i, \quad \tau_{i-1} < t \leq \tau_i; \quad i = 1, \dots, N + 1,$$

where we define the set of change points as an ordered collection of points in $\{1, \dots, n\}$,

$$CP := \{\tau_i : 1 < \tau_1 < \tau_2 < \dots < \tau_N < n\}$$

with the convention $\tau_0 = 0; \tau_{N+1} = n$.





(a) Measurements of the annual flow of the river Nile at Ashwan 1871-1970. (b) Yearly counts of British coal-mining disasters between 1851-1962.

Epidemic change point

- A special type of multiple change-point problems is the so-called epidemic change-point problem:

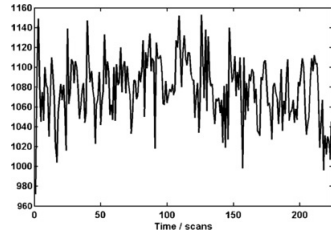
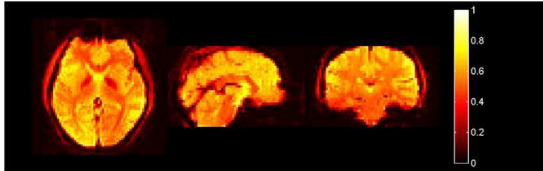
we can view a process $Y = \{Y_t : t = 1, \dots, n\}$ as following some distribution P_1 between the two time points τ_1 and τ_2 and another distribution within the rest time:

$$Y_t \sim P_1, \quad \tau_1 < t \leq \tau_2,$$

and

$$Y_t \sim P_2, \quad t \in \{1, \dots, \tau_1, \tau_2 + 1, \dots, n\}.$$

- The period $(\tau_1, \tau_2]$ is interpreted as an *epidemic deviation* from the usual state of the process Y and $\tau^* := \tau_2 - \tau_1$ is called the *duration of the epidemic state*.



- ▶ Example of fMRI data set for a single subject from the 1000 connectors. The left image shows a three-dimensional view of the temporally averaged brain data and is formally equivalent to the mean function for these data.
- ▶ The right image shows the time series for the central voxel of the image.

Off-line vs on-line setting

- ▶ When observations $X_1, \dots, X_n$ are obtained one at a time, we say that the setting is online or sequential.
 - ▶ The goal is to detect changes as quickly as possible while keeping the number of false alarms as low as possible.
- ▶ We talk about off-line or a posteriori detection of changes when the data is obtained all at once.
 - ▶ In this case, there is no need for real-time processing of the data, and one can take advantage of this additional computation time.

Single vs multiple change-point detection

- ▶ There is quite a difference between detecting a single change-point and multiple change-points in the off-line setting.
- ▶ It is important to understand that the second problem is much harder since the number of possible outcomes for the procedure jumps from $n - 1$ to 2^{n-1} if the number of change-points is unknown, where n is the number of observations.
 - ▶ There are $\binom{n-1}{k-1}$ ways for a signal of size n to be segmented in k segments, hence a total of $\sum_{k=1}^{n-1} \binom{n-1}{k-1} = 2^{n-1}$ possibilities.

Parametric vs non-parametric

- ▶ It is common to make parametric assumptions, i.e., to assume that the distribution of the observations belongs to a family of distributions that can be described using a finite number of parameters.
- ▶ Thus, the changes occur in one (or more) of the distribution parameters.
- ▶ These assumptions are important for the design and the analysis of change-point procedures.

Detection of changes in the mean function

- Assume that the functional observations

$$X_i = (X_i(t), t \in \mathcal{I}) \in L_2(\mathcal{I}), \quad 1 \leq i \leq n$$

are independent, and we want to test if their mean remains constant in i .

- To this aim, we consider the model

$$X_i(t) = \begin{cases} \mu_1(t) + Y_i(t), & 1 \leq i \leq k^*; \\ \mu_2(t) + Y_i(t), & k^* < i \leq n; \end{cases}$$

in which the

$$Y_i(t) \stackrel{iid}{\sim} Sp(0, \gamma);$$

the mean functions μ_1 and μ_2 are in $L_2(\mathcal{I})$, and we assume that

$$k^* = \lceil n\theta^* \rceil \quad \text{for some } 0 < \theta^* < 1.$$

- It can be shown that the sample covariance of the functional observations converge to the covariance function

$$\tilde{\gamma}(t; s) = \gamma(t, s) + \theta(1 - \theta)(\mu_1(t) - \mu_2(t))(\mu_1(s) - \mu_2(s)).$$

- It has orthonormal eigenfunctions w_k and nonnegative eigenvalues λ_k satisfying

$$\int_0^1 \tilde{\gamma}(t, s) w_k(s) ds = \lambda_k w_k(t).$$

- The quantities $\tilde{\gamma}(t, s)$, w_k and λ_k are used to describe the distribution of the test statistic under the alternative of a single change point.

Detection procedure

- To explain the idea of the test procedure, denote

$$\hat{\mu}_{1,k}(t) = \frac{1}{k} \sum_{i=1}^k X_i(t), \quad \hat{\mu}_{k,n} = \frac{1}{n-k} \sum_{i=k+1}^n X_i(t).$$

- If the mean is constant, the difference

$$\Delta_{n,k}(t) := |\hat{\mu}_{1,k}(t) - \hat{\mu}_{k,n}(t)|$$

is small for all $1 \leq k < n$ and all $t \in \mathcal{I}$.

- However, $\Delta_{n,k}(t)$ can become large due to chance variability if k is close to 1 or to n .
- It is, therefore, usual to work with the sequence

$$P_{n,k}(t) = \frac{k(n-k)}{n} \Delta_{n,k}(t) = \left| \sum_{i=1}^k [X_i(t) - \bar{X}_n(t)] \right|,$$

in which a parabolic weight function attenuates the variability at the endpoints.

- If we new that $k^* = k$, then the measure of heterogeneity could be

$$P_{n,k}(t) = \left| \sum_{i=1}^k [X_i(t) - \bar{X}_n(t)] \right|$$

or its L_2 version

$$D_{n,k} = \int_0^1 P_{n,k}^2(t) dt = \int_0^1 \left| \sum_{i=1}^k [X_i(t) - \bar{X}_n(t)] \right|^2 dt.$$

If k^* is unknown, then we consider the corresponding maximal statistics:

- point-wise

$$P_n(t) = \max_{1 \leq k < n} P_{n,k}(t);$$

- its L_2 -version

$$D_n := \max_{1 \leq k < n} D_{n,k} = \max_{1 \leq k < n} \int_0^1 \left| \sum_{i=1}^k [X_i(t) - \bar{X}_n(t)] \right|^2 dt.$$

So we arrive at a problem to solve

$$P(D_n > C | H_0) = \alpha.$$

We shall modify the test statistics.

- ▶ Let $\lambda_1 > \lambda_2 > \dots > \lambda_d$ be the largest d eigenvalues of the covariance $\gamma(t, s)$.
- ▶ Let $v_1(t), v_2(t), \dots, v_d(t)$ be the corresponding eigenfunctions:

$$\int_0^1 \gamma(t, s) v_k(s) ds = \lambda_k v_k(t).$$

- ▶ Recall that the estimated eigenelements are defined by

$$\int_0^1 \hat{\gamma}(t, s) \hat{v}_k(s) ds = \hat{\lambda}_k \hat{v}_k(t).$$

where

$$\hat{\gamma}(t, s) = \frac{1}{n} \sum_{i=1}^n (X_i(t) - \bar{X}_n(t))(X_i(s) - \bar{X}_n(s)).$$

- Consider the scores corresponding to the largest d eigenvalues:

$$\hat{\xi}_{\ell,i} = \int_0^1 (X_i(t) - \bar{X}_n(t)) \hat{v}_{\ell}(t) dt, \quad i = 1, 2, \dots, n; \quad \ell = 1, 2, \dots, d.$$

- Introduce the statistic

$$T_n(d) = \sum_{k=1}^n T_{n,k}(d),$$

where

$$T_{nk}(d) = \sum_{\ell=1}^d \hat{\lambda}_{\ell}^{-1} \left[\sum_{i=1}^k \hat{\xi}_{\ell,i} - \frac{k}{n} \sum_{j=1}^n \hat{\xi}_{\ell,j} \right]^2.$$

Theorem

Suppose Assumptions

(1) $\mu_1, \mu_2 \in L_2$ and (Y_i) are iid in L_2 ;

(2) The eigenvalues λ_i satisfy, for some $d > 0$

$$\lambda_1 < \lambda_2 < \cdots < \lambda_d < \lambda_{d+1};$$

(3) $\int E(Y_i^4(t)) dt < \infty$

hold. Then, under H_0 ,

$$\lim_{n \rightarrow \infty} P(n^{-2} T_n(d) > x) = P(T(d) > x),$$

where

$$T(d) := \int_0^1 \sum_{\ell=1}^d B_\ell^2(u) du,$$

where $B_1, \dots, B_d$ denote independent standard Brownian bridges.

- ▶ The distribution of the random variable $T(d)$ was derived by Kiefer in 1959.
- ▶ Denoting by $c_d(\alpha)$ its $(1 - \alpha)$ 'th quantile, the test rejects H_0 if $T_n(d) > c_d(\alpha)$.

Nominal size	d					
	1	2	3	4	5	6
10%	0.345165	0.606783	0.842567	1.065349	1.279713	1.485200
5%	0.460496	0.748785	1.001390	1.239675	1.469008	1.684729
1%	0.740138	1.072101	1.352099	1.626695	1.866702	2.125950
	7	8	9	10	11	12
10%	1.690773	1.897365	2.096615	2.288572	2.496635	2.686238
5%	1.895557	2.124153	2.322674	2.526781	2.744438	2.949004
1%	2.342252	2.589244	2.809778	3.033944	3.268031	3.491102
	13	14	15	16	17	18
10%	2.884214	3.066906	3.268958	3.462039	3.650724	3.837678
5%	3.147604	3.336262	3.544633	3.740248	3.949054	4.136169
1%	3.708033	3.903995	4.116829	4.317087	4.554650	4.734714
	19	20	21	22	23	24
10%	4.024313	4.214800	4.404677	4.591972	4.778715	4.965613
5%	4.327286	4.532917	4.718904	4.908332	5.101896	5.303462
1%	4.974172	5.156282	5.369309	5.576596	5.759427	5.973941
	25	26	27	28	29	30
10%	5.159057	5.346543	5.521107	5.714145	5.885108	6.083306
5%	5.495721	5.688849	5.866095	6.068351	6.242770	6.444772
1%	6.203718	6.393582	6.572949	6.771058	6.977607	7.186491

Simulated critical values of the distribution of $T(d)$.